

BRASEN XU

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EDUCATION

University of Waterloo

Sep 2022 - Apr 2027

Bachelor of Computer Science (Co-op), AI Specialization, **3.8 GPA, 87.5 CAV**

EXPERIENCE

Splunk

May 2025 - August 2025

Software Engineer Intern

San Francisco, CA

- Developed scalable Java Spring Boot REST APIs and UI workflows for on-call incident response workflows, reducing investigation time by **30x** on average.
- Built Java Spring microservices for runtime configuration of AWS ML pipelines, reducing parameter change deployment time by **99%**.
- Built self-service API endpoints for AWS Glue job execution, cutting operational time by **93% (30 mins to 2 mins)** and eliminating manual console workflows.
- Resolved flaky integration tests by fixing race conditions and enhancing Mockito/JUnit mocks, reducing CI/CD pipeline runtime by **75% (7 hours to 1-2 hours)**.
- Awarded **Most Impactful Intern Project** in Security Org for achieving the term's top efficiency gains.

Onsemi

September 2024 - December 2024

Software Engineer Intern

Waterloo, ON

- Built a language compiler using ANTLR4, reducing compilation time by **30%** and enabling the generation of **50+ core microcode configuration files**.
- Developed configuration UIs for **5+ semiconductor products** using the QT framework and C++, enabling intuitive parameter tuning and real-time performance monitoring for manufacturing operators.
- Migrated legacy Python 2 processing core to Python 3, improving execution speed by **20%** and enabling integration with modern frameworks.

Siemens

January 2024 - April 2024

Software Engineer Intern

Toronto, ON

- Developed a C# display application with SQL Server backend for real-time parts tracking, reducing inventory lookup time by **96%** on average.
- Built a Windows Form application to streamline PCBA assembly workflows across **11 product lines**, reducing overall assembly errors by **30%**.

Insurance Bureau of Canada

May 2023 - August 2023

Software Engineer Intern

Toronto, ON

- Created automated test scripts for manual test cases using Selenium and Katalon Studio, boosting efficiency and reducing overall testing times by **over 95% (20x)**.

PROJECTS

MotiSpectra - ConUHacks 2nd Place

- Built a real-time emotion and attentiveness analysis app using **Next.js**, **Python**, and custom **ML models**, integrating into Zoom/Meet/Teams and visualizing results via dynamic radar graphs with **97%** emotion detection accuracy.

NarratorRL - MetHacks 1st Place

- Created a **React Native** app for vision-impaired users that uses **Tesseract OCR**, **Django**, and **Cohere NLP** to read text aloud, summarize content, detect language, and verify legibility in real time.

SKILLS

Languages

Java, Python, C++, JavaScript, TypeScript, C#, SQL, Bash, C, Kotlin, R, HTML, CSS

Frameworks

Spring Boot, React, Next.js, Node.js, Django, .NET, QT, Ruby on Rails, FastAPI

Tools

AWS, Git, PostgreSQL, MongoDB, Kubernetes, Linux, Docker, Azure, Jira, Jupyter

Technologies

CI/CD, Distributed Systems, System Design, OOP, Data Structures & Algorithms, Machine Learning